

Exercise 3-Student marks Processing

```
206000airflow@137ba4301055:/opt/airflow/dags$ cd /opt/airflow/dags
airflow@137ba4301055:/opt/airflow/dags$ cat > exercise3.py
from airflow import DAG
from airflow.operators.python import PythonOperator
from datetime import datetime

def create_marks_file():
    with open('/tmp/students.txt', 'w') as f:
        f.write(
            "Rahul,85\n"
            "Priya,92\n"
            "Amit,78\n"
            "Sneha,95"
        )
```

```
task1-77 task2-77 task3
airflow@137ba4301055:/opt/airflow/dags$ python exercise3.py
2026-06-23T09:58:35.857211Z [warning] The `airflow.operators.python.PythonOperator` attribute is deprecated. Please use `airflow.operators.python.PythonOperator`. [py.warnings] category=DeprecatedImportWarning filename=/opt/airflow/dags/exercise3.py
airflow@137ba4301055:/opt/airflow/dags$ airflow dags reserialize
2026-06-23T10:00:00.341648Z [info] ] setup plugin alembic.autogenerate.schemas [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T10:00:00.341883Z [info] ] setup plugin alembic.autogenerate.tables [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T10:00:00.342016Z [info] ] setup plugin alembic.autogenerate.types [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T10:00:00.342124Z [info] ] setup plugin alembic.autogenerate.constraints [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T10:00:00.342279Z [info] ] setup plugin alembic.autogenerate.defaults [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T10:00:00.342506Z [info] ] setup plugin alembic.autogenerate.comments [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T10:00:01.244602Z [info] ] DAG bundles loaded: dags-folder [airflow.dag_processing.bundles.manager.DagBundlesManager]
2026-06-23T10:00:01.257240Z [info] ] Filling up the DagBag from /opt/airflow/dags [airflow.dag_processing.dagbag.DagBag]
```

```
airflow@137ba4301055:/opt/airflow/dags$ airflow dags list
2026-06-23T10:00:11.716570Z [info] ] setup plugin alembic.autogenerate.schemas [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T10:00:11.718147Z [info] ] setup plugin alembic.autogenerate.tables [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T10:00:11.719058Z [info] ] setup plugin alembic.autogenerate.types [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T10:00:11.720155Z [info] ] setup plugin alembic.autogenerate.constraints [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T10:00:11.720764Z [info] ] setup plugin alembic.autogenerate.defaults [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T10:00:11.721683Z [info] ] setup plugin alembic.autogenerate.comments [alembic.runtime.plugins] loc=plugins.py:37
dag_id | fileloc | owners | is_paused | bundle_name | bundle_version
-----|-----|-----|-----|-----|-----
exercise1_file_read | /opt/airflow/dags/exercise1.py | airflow | False | dags-folder | None
exercise2_salary_report | /opt/airflow/dags/exercise2.py | airflow | False | dags-folder | None
exercise3_student_result | /opt/airflow/dags/exercise3.py | airflow | True | dags-folder | None
```

```
airflow@efe21df28c32:/opt/airflow/dags$ cat /tmp/result.txt
Student Result Report
Students = 4
Average Marks = 87.5airflow@efe21df28c32:/opt/airflow/dags$ cat /tmp/students.txt
Rahul,85
Priya,92
Amit,78
Sneha,95airflow@efe21df28c32:/opt/airflow/dags$ cat /tmp/result.txt
Student Result Report
Students = 4
Average Marks = 87.5airflow@efe21df28c32:/opt/airflow/dags$
```

Dag

exercise3_student_result

manual_2026-06-23T10:16:06.501511+00:00

Home

Dags

Assets

Browse

Admin

Docs

00:00:05.438

00:00:02.719

create_marks_file

calculate_average

generate_result

Search Dags Ctrl+K

Trigger

manual_2026-06-23T10:16:06.501511+00:00

Success

Logical Date

Run Type

Start Date

End Date

Duration

Triggering User

Dag Version(s)

2026-06-23 15:46:04

Manual

2026-06-23 15:46:07

2026-06-23 15:46:12

00:00:05.438

admin

v1

Task Instances

Asset Events

Audit Log

Code

Details

+ Filter

Task ID	Map Index	State	Start Date	Try Number	Operator	Duration	
generate_result		Success	2026-06-23 15:46:11	1	PythonOperator	00:00:00.087	✓ / ✕
calculate_average		Success	2026-06-23 15:46:10	1	PythonOperator	00:00:00.087	✓ / ✕
create_marks_file		Success	2026-06-23 15:46:07	1	PythonOperator	00:00:05.550	✓ / ✕